High circle Transform

1. Calculate radius for each combination

Formula:

$$r = \sqrt{(x-x_c)^2+(y-y_c)^2}$$

Given:

$P1 = (5,4), P_2 = (7,6)$

$C1 = (2,4), C_2 = (4,3)$

P_1 with $C1$:

$$r = \sqrt{(5-2)^2+(4-4)^2}$$

$$= \sqrt{9} = 3$$

$P1$ with $C2$:

$$r = \sqrt{(5-4)^2+(4-3)^2}$$

$$= \sqrt{1+1} = \sqrt{2} \simeq 1.41$$

P_2 with $C1$:

$$r = \sqrt{(7-2)^2+(6-4)^2}$$

$$= \sqrt{25+4} = \sqrt{29} \approx 5.39$$

P_2 with $C2$:

$$r = \sqrt{(7-4)^2+(6-3)^2}$$

$$= \sqrt{9+9} = \sqrt{18} \simeq 4.24$$

C1 gets maximum matching radius value for the given points: $r=3$ and 5.39.

Given $P(8,6)$, $C(4,3)$

Formula:

Formula:

$$r = \sqrt{(x - x_c)^2 + (y - y_c)^2}$$

$$r = \sqrt{(8-4)^2 + (6-3)^2}$$

$$= \sqrt{(4^2 + 3^2)}$$

$$= \sqrt{16+9} = \sqrt{25}$$

$$r = 5$$

(3) Given: co

Center, $C = (5,4)$

Edge point, $P = (8,8)$

Formula:

$$r = \sqrt{(x - x_c)^2 + (y - y_c)^2}$$

Substitute the values:

$$r = \sqrt{(8-5)^2 + (8-4)^2}$$

$$= \sqrt{3^2 + 4^2}$$

$$= \sqrt{9+16}$$

$$= \sqrt{25}$$

$$r = 5$$

The radius of the circle is 5.